Background

I am playing in my Fantasy Basketball league with little knowledge of basketball. However, I am a Statistics major at the very lovely University of Chicago, and I do really like the subject, so I'm going to undergo building a statistical model to inform my strategy for when to lock players! If you guys are unfamiliar with the Sleeper Fantasy lock-in mode rules, you can read them here: ta-da

Setup

On every team, and on my team, there are 5 types of positions, and 9 slots for players.

The 9 slots

Slot	**Eligible Positions**
PG	PG
SG	SG
PF	PF
SF	SF
C	C
G	PG, SG
F	PF, SF
UTIL	PG, SG, PF, SF, C
UTIL	PG, SG, PF, SF, C

Players

Each player *i* has the following attributes:

- One or more positions
- N_i games a week
- An expected number $E[P_{i,j}]$ points at game j
- We assume for simplicity, and from sports precedent, that each

*But that this expected number of points/game can have some kind of unknown error, and that across the course of the season I want my model to uncover this error and give me a better version of this but anyways that's somewhere down the line

For instance, we might have the following player Lebron James, who is eligible for three slots! PG, SG, and SF positions:

$E[P_{James,\ 1}]$	$E[P_{James,\ 2}]$	$E[P_{James,\ 3}]$	$E[P_{James,\ 4}]$
31.96	35.43	35.33	32.41

Motivating / Simple Case

Let's say we have two players, Devin Booker and Ja Morant, who are the only two eligible players that a team has for shooting guard; yes unfortunately I did not draft carefully.

Devin Booker

$E[P_{Booker,\ 1}]$	$E[P_{Booker,\ 2}]$
40.1	40.2

Ja Morant

$E[P_{Booker,\ 1}]$	$E[P_{Booker,\ 2}]$
29.2	29.2

Our dilemma is: whether we should lock the players after their day one performances, and how should we decide that? This is probably dependent on how everyone else is contributing, but for simplicity let's just treat this like an isolated case.

Working with the extremes?

The extremes: if Devin gets 50.1 points, we definitely should lock. Basically, we are saying conditional on some points by Ja Morant and Devin Booker in the "first round" of games, what's the probability that we are able to exceed that number in the next two games (or draws from this random distribution?) and framed like this, it's a very simple problem.

We want the probability that when we draw from Devin Booker's distribution, and then draw again from Ja Morant's distribution, that the maximum of those two exceeds our current score. If that probability exceeds 50% then we lock, and if that probability is less than 50% then we don't draw. Basically, we should take $E[\max(P_{Booker,\ 2}, P_{Morant,\ 2})]$ and if this value exceeds our current max score, then we should not lock, and if this value does not exceed our current max score, then we shouldn't.

Extending this case to a simple general model

Actually, going about this model, it's a pretty simple problem. We can basically extend this case in two ways, first with the number of games and second with the number of slots.

First, let's say Booker and Morant have 3 games instead of 2. Then, our method for deciding whether to lock is simply just:

$$E[\max(P_{Booker,\ 2}, P_{Morant,\ 2}, P_{Booker,\ 3}, P_{Morant,\ 3})]$$

Second, let's say for simplicity that there are two SG slots instead of one, and there is a third player Paul George who has two games this week. Then, we decide to lock by:

1. Drawing from each of these distributions
2. Recording the maximum and second maximum
3. And finding the expected maximum and second maximum

$$m_1, m_2 = E[\max_2(P_{Booker,\,2}, P_{Morant,\,2}, P_{Booker,\,3}, P_{Morant,\,3})]$$

The notation is iffy, but the idea is that we are going to get up to k maximums, where k is the number of slots that are available.

Should we actually lock I

There is an edge case, where I'll take the extremes of, to kind of illustrate the dilemma that I'm in, which actually makes me think this is a Markov chain? Anyways here's the situation.

Let's set an extreme example where there are 2 slots, and 3 people. Paul George performs incredibly well, and so say we just kind of want to hold him out:

Devin Booker

$E[P_{Booker,\,1}]$	$E[P_{Booker,\,2}]$
40.1	40.2

Ja Morant

$E[P_{Booker,\,1}]$	$E[P_{Booker,\,2}]$
29.2	29.2

Paul George

$E[P_{George,\,1}]$	$E[P_{Booker,\,2}]$
80	90

Let's say Ja Morant scores like well above his expected, scoring a 40, and Devin Booker scores a 30, and like an anomaly, Paul George scores a 20. What do we do here? My intuition is that it seems like locking Ja Morant allows us to draw from a significantly more advantageous distribution, which is one that involves both Devin Booker. So we are effectively comparing two cases:

1. Case 1, where we do lock Ja Morant:
    a. Total Score = $40 + E[\max(P_{Booker,\,2}, P_{George,\,2})]$
2. Case 2, where we don't lock Ja Morant:
    a. Total Score = $E[\max_2(P_{Booker,\,2}, P_{George,\,2}, P_{Morant,\,2})]$

So which one is larger? I am not sure. But I think we're getting to some kind of heuristic, which is that maybe it's better if one of your low EV people performs well, which lets you draw randomly from your higher EV people? Anyways let's extend this to an even more complex case I think because there is an intersting point that is being illustrated here!

A more complex case

Let's say these are the players (now in more concise table form) and there are three slots available; then there are ${}_5C_3$ + 2 (4 players + 1 blank = 5, all locks and

Player	$E[P_{player,\ 1}]$	$P_{player,\ 1}$	$E[P_{player,\ 2}]$	$P_{player,\ 2}$
Devin Booker	40	**40**	40	??
Ja Morant	30	**45**	30	??
Paul George	70	**60**	70	??
Anthony Edwards	80	**70**	80	??

We want to maximize the function of the expected value of the next distribution over each of our 11 lock cases. Is this a DP problem with probabilities? Or a Markov Chain problem? Anyways, in the following cases below, we want to maximize over the expected max total points function.

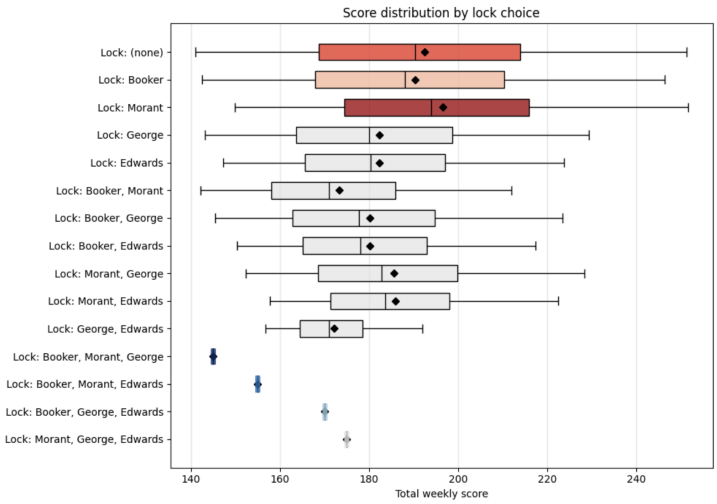
Locks	Number of unique cases	Expected max total points
No Locks	1	$E[\max_3\{P_{player,\ 2}\}]$
One lock, player a	4	$P_{a,\ 1} + E[\max_2\{P_{player,\ 2} : player \neq a\}]$
Two locks, players i and players j	6	$P_{a,\ 1} + P_{b,\ 1} + E[\max\{P_{player,\ 2} : player \neq a, b\}]$
3 locks	4	$P_{i,\ 1} + P_{j,\ 1} + P_{k,\ 1}$

Finding the expected total score and quartiles using Monte Carlo Simulation

The easiest approach for us to recover the mean/quarties is via simple Monte Carlo Simulation, where we just sample from each player's remaining game Gamma distribution, and then find the average of the m_1 and m_2 which suffices for our current situation. We've (Claude) coded it and we can find the code in the appendix/Git/wherever we've decided to store this.

Anyways the preliminary results of this are below and this is so awesome holy fucking shit. We first have the table of the expected score given some of our lock cases, and then next the box plot which gives us the quartile / 95th CI for where our weekly total score is going to land. Incredibly cool stuff!

Booker	Morant	George	Edwards	Expected score
				192.29
40				190.30
	45			196.41
		60		182.21
			70	182.20
40	45			173.18
40		60		180.14
40			70	180.18
	45	60		185.57
	45		70	185.97
		60	70	172.12
40	45	60		145.00
40	45		70	155.00
40		60	70	170.00
	45	60	70	175.00



Where are we right now (Checkpoint #1)

Right now, we've put together a pretty rigorous mechanism for where:

- given N players with K slots
- and each of these N players has 1 game remaining
- and given we have all the scores of the current round

We can best decide who to lock, and who not to lock and roll the dice on for the next round! This is cool, but there are some next steps that I would like to traverse, because we're still a little far off from where we need to be. I'm going to unpack some of the assumptions we made, so that we can further refine our 1-week model.

1. We assumed that all the games happen at once. In reality, games happen in staggered increments, so there might not exist some point in time where all $P_{i,j}$ happen before all $P_{i,j+1}$
2. The slots are valued equally. These slots are actually not valued equally, since the set of players that can enter slot PG is a subset of the set of players that can enter slot G which is a subset of players that can enter slot UTIL

Differently Valued Slots / Complex Case I

Based on my current situation, let's say there are two open slots, one is the SF slot and one is the F slot. The following two players are eligible for the SF slot:

Player	$E[P_{player,1}]$	$P_{player,1}$	$E[P_{player,2}]$	$P_{player,2}$
Lebron James	32	40	35	??
Jayson Tatum	42	33	42	??

In the broader F slot, the following four players are eligible for the F slot:

Player	$E[P_{player,\ 1}]$	$P_{player,\ 1}$	$E[P_{player,\ 2}]$	$P_{player,\ 2}$
Lebron James	32	**40**	35	??
Jayson Tatum	42	**33**	42	??
Paskal Siakam	37	**42**	37	??
Chet Holmgren	36	**37**	37	??

This changes things a little bit. Let's think about the differents kinds of cases we might have, given we have two non-equally valued slots:

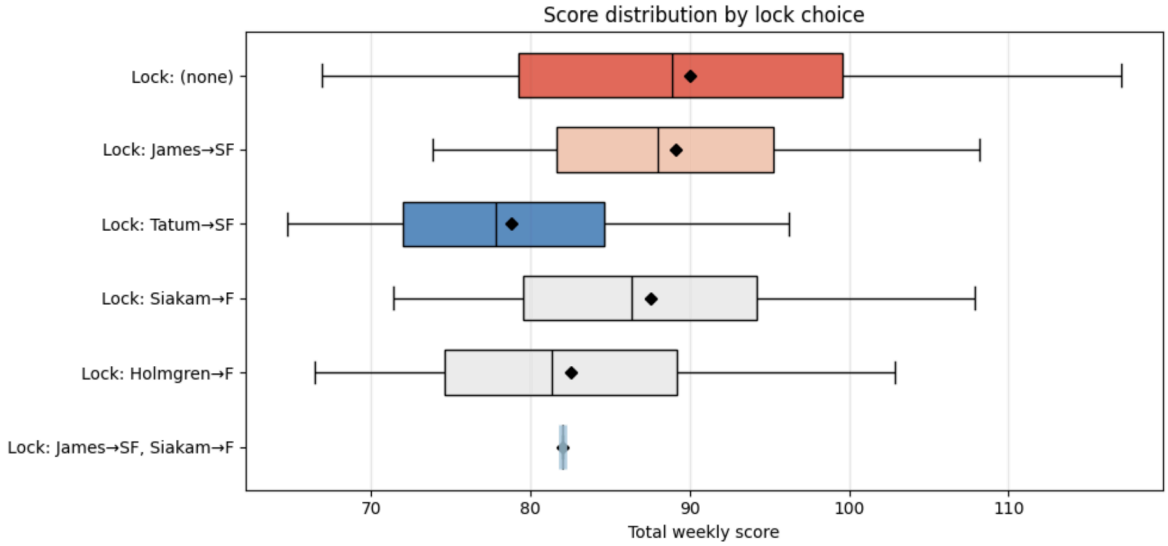
Locks	Number of unique cases	Expected max total points
No Locks	1	$E[max_2\{P_{player,\ 2} : \text{one of the players is an SF}\}]$
One SF lock, player a *No F locks*	${}_2C_1 * 1 = 2$	$P_{A,\ 1} + E[max_1\{P_{player,\ 2} : \text{player} \neq a\}]$
No SF lock *One F lock, non-SF player C*	$1 * {}_2C_1 = 2$	$P_{C,\ 1} + E[max_1\{P_{player,\ 2} : \text{player is SF}\}]$ since there are three SFs, we will always draw one
No SF lock *One F lock, SF player A*	This case is actually redundant, because in no situation would we ever slot the SF player in the general F slot as opposed to the more precise SF slot	
One SF lock *One F lock*	1	$\max(P_{a,\ 1} + P_{c,\ 1})$ this was changed from the previous label to save visualization space

Again, we can find the maximum of these values using Monte Carlo Simulation.

- For the no locks case, we can just take the largest SF score from the 2 fielded candidates, and then from the remaining 3 candidates take the maximum there.
- For both the one locks cases, we take the same approach as the simple Monte Carlo earlier, and draw the top two of the remaining three players
- For the two locks cases, we simply just take the maximum of the two SFs, and then the maximum of the remaining three

This actually resolves a large chunk of the puzzle, because we only ever have one slot anyways, except the UTIL case and SF/PF or even the SF/PG cases which I think are a bit more complicated, but will mostly just end up being sampling again? Anyways, let's see empirically if this makes sense:

James	Tatum	Siakam	Holmgren	Expected score
				89.96
40				**89.03**
	33			**78.80**
		42		87.52
			37	82.52
40		**42**		**82.00**



And it does! It looks super cool, awesome, aligned with intuition, and shows that actually the no lock is a little higher risk, higher reward, and would kind of be affected on where and how I am doing as a team.

Finally, let's try a slightly more general case, where there are h F-slots and 1 specialized SF slot, as well as x number of SF players in set of SF players A, and y number of F players in set B:

Locks	Number of unique cases	Expected max total points
No SF lock *m F locks, all non-SF players* *where* $0 \leq m \leq min(h, y)$	- -	$\Sigma P_{m,1} + E[\max_1\{P_{player,2} : player \in A\}] +$ $E[\max_{h-m}\{P_{player,2} : player \neq a \text{ and } player \notin M\}]$ where a is the max SF selected in the first part
One SF lock, player a in A *m F locks where* $0 \leq m \leq h$	- -	$P_{A,1} + \Sigma P_{m,1} + E[\max_{h-m}\{P_{player,2} : player \neq a \text{ and } player \notin M\}]$

As always, we can find the maximum of these values using Monte Carlo Simulation.

- For the no SF lock, m F lock case, we can just take the maximum SF of each draw, and then take the next h-m maximums of the corresponding draw
- For the one SF lock, we can just take the maximum h-m of the draw we make, and fill the unlocked spots out accordingly. This is also quite simple

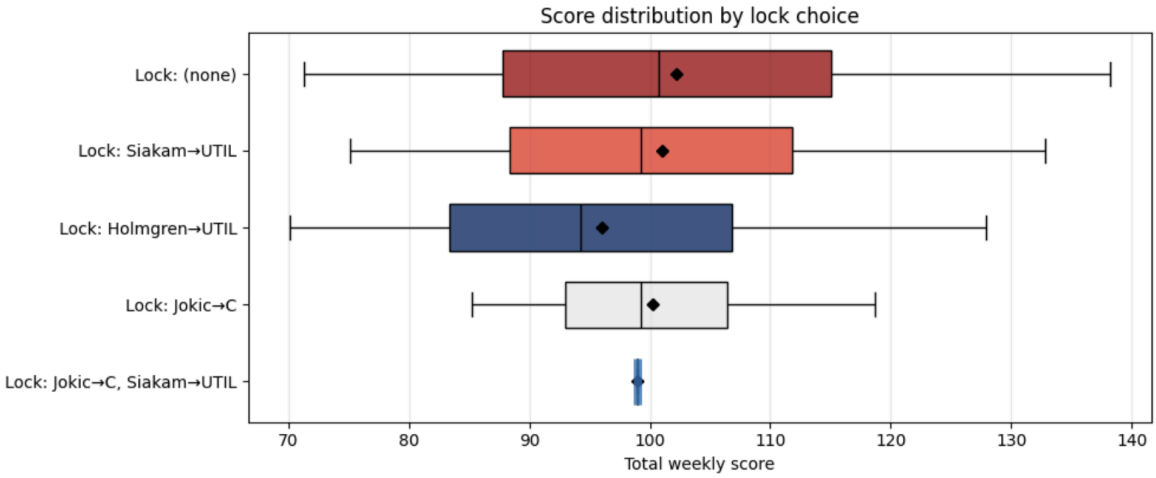
These cases are heightened by the fact that we need to do these for all values of m, and at all values of m, we have to pick the corresponding m F locks. So maybe this is computationally intensive, but since h is always ≤ 3 then I think the computational-ness of it be relatively cheap. Anyways, let's deal with this empirically:

A few examples of this in motion!

For our center Nikola Jokic, we ideally want him to take the C spot, but we are also open to "carving out" a UTIL spot for the other two competing C in Holmgren and Siakam. So, we run this simulation where h = 1, and the specialized spot is C, and we get the following interesting and compelling results:

This model would actually be simulating the correct circumstance as of Thursday evening, because each of my centers would have one game remaining.

Jokic	Siakam	Holmgren	Expected score
			102.19
	42 (UTIL)		100.99
		37 (UTIL)	95.99
57 (C)			100.20
57 (C)	42 (UTIL)		99.00



The takeaway for now is that if the best combination involves leaving the UTIL spot free, then I should add more weight to it since the UTIL spot can also be filled by other stuff; basically they assume that my combination of Jokic/Siakam/Holmgren in their final remaining game can be distributed across the C and UTIL spot better than if I locked right now. Which is super interesting and very good to know.